

ADAPTIVE SILENT COMMUNICATION SYSTEM USING VIBRATION PATTERN RECOGNITION

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Abstract— Communication plays a vital role in expressing needs, emotions, and emergency requirements. However, speech-impaired individuals, elderly people, and patients with limited mobility often face difficulties in communicating effectively with caregivers. Conventional communication methods such as verbal interaction, writing, and mobile devices are not always practical during emergency situations. This paper presents an **Adaptive Silent Communication System using ESP32 and ESP-NOW Protocol** to provide a simple, reliable, and wireless assistive communication solution. The proposed system consists of two ESP32 microcontrollers functioning as transmitter and receiver units. A single push button is used to generate communication patterns through short and long button presses. The generated patterns are transmitted wirelessly using the ESP-NOW protocol without requiring internet connectivity or a Wi-Fi router. The receiver compares the received communication pattern with predefined patterns stored in memory. When a valid pattern is recognized, the corresponding message is displayed on an SSD1306 OLED display, and a vibration motor provides tactile feedback confirming successful communication. Invalid patterns are also detected and displayed appropriately. The proposed system offers low power consumption, portability, low implementation cost, and reliable wireless communication, making it suitable for healthcare, rehabilitation centers, elderly care, and assistive communication applications.

Keywords— Adaptive Silent Communication, ESP32, ESP-NOW, Pattern Recognition, OLED Display, Vibration Motor, Embedded Systems, Assistive Technology.

I. INTRODUCTION

The advancement of embedded systems and wireless communication technologies has enabled the development of compact and cost-effective communication solutions. ESP32-based platforms provide the processing and wireless communication capabilities required for a wide range of embedded and IoT applications [1], [2]. The ESP-NOW protocol further supports direct communication between ESP32 devices without requiring a conventional Wi-Fi network or internet connectivity [3].

This project focuses on developing a silent communication approach based on predefined vibration patterns. Short and long button inputs are converted into corresponding patterns and transmitted wirelessly from a transmitter to a receiver. The receiving device processes the transmitted pattern and

identifies the corresponding predefined message.

An SSD1306 OLED display is used to provide visual information to the user, while a vibration motor provides tactile feedback. The integration of embedded processing, wireless communication, and vibration-based interaction provides a simple and low-cost approach to silent communication [4], [7].

The developed system was implemented and experimentally tested using predefined patterns such as S, L, and SL. The results demonstrate successful pattern transmission, recognition, message display, and vibration feedback.

An SSD1306 OLED display is used to display the recognized message, while a vibration motor provides tactile feedback. The system combines embedded programming, wireless communication, and vibration-based interaction to provide a simple communication method [4], [7].

The system was implemented and tested using predefined patterns such as S, L, and SL. The results demonstrate successful pattern transmission, recognition, message display, and vibration feedback.

II. LITERATURE SURVEY

Recent developments in embedded and wireless communication systems have enabled the design of compact and low-cost communication devices. ESP32 provides integrated processing and wireless communication capabilities, making it suitable for embedded and IoT-based applications [1], [2]. The availability of ESP-NOW further supports direct communication between ESP32 devices, providing a lightweight method for exchanging data without depending on a conventional Wi-Fi network [3].

Embedded software frameworks provide the necessary interfaces for implementing communication, input processing, and device control. ESP-IDF offers development support for ESP32-based applications and facilitates the implementation of embedded communication systems [4]. In addition, embedded-system development approaches emphasize efficient hardware and software integration for reliable real-time operation [12], [13].

Wireless communication has also been explored in assistive and healthcare-oriented systems, where low-cost embedded devices can provide alternative methods of information exchange [17]–[19]. Such systems demonstrate the potential of combining wireless communication with embedded hardware to support users in situations where conventional communication methods may be inconvenient.

Display and feedback mechanisms are important for

communicating the recognized information to the user. The SSD1306 controller provides a compact OLED display interface suitable for embedded applications [7]. Libraries and development tools further simplify the integration of OLED displays with microcontroller-based systems [9].

The reviewed literature and technical references indicate that ESP32-based embedded systems, wireless communication protocols, and compact user-feedback devices can be effectively integrated to develop practical communication solutions. However, the proposed system focuses specifically on combining predefined short and long vibration patterns with ESP-NOW communication, OLED message display, and vibration feedback to provide a simple silent communication approach.

III. PROBLEM STATEMENT

Communication is one of the fundamental requirements for individuals to express their needs and request assistance. However, speech-impaired individuals, elderly people, and patients with severe medical conditions may experience difficulties in communicating effectively. Conventional communication methods based on verbal interaction, writing, or mobile applications may not always be practical, particularly during emergency situations or for users with limited physical abilities [17]–[19].

The major problems addressed by the proposed system are summarized below.

A. Communication Difficulties

- Speech-impaired individuals and patients may face challenges in conveying their requirements quickly and effectively [17], [19].
- Traditional communication methods may not always be suitable during emergency situations.

B. Dependence on Verbal Communication

- Many conventional communication methods rely on speech writing.
- These methods may become ineffective for users with communication difficulties [17]–[19].

C. Requirement of Internet Connectivity

- Several wireless communication systems depend on Wi-Fi infrastructure or internet connectivity.
- Communication may become difficult in locations where suitable network infrastructure is unavailable. ESP-NOW provides direct communication between ESP32 devices without requiring a conventional Wi-Fi network [3].

D. Complex User Interfaces

- Existing assistive communication devices may employ multiple buttons, touchscreens, or mobile applications.
 - Such interfaces can increase system complexity and may make operation difficult for elderly or physically challenged users.

E. Lack of Reliable Wireless Communication

- Some communication systems require additional communication modules or network infrastructure.
- This can increase implementation cost and system complexity. ESP32 provides integrated wireless

communication capabilities suitable for compact embedded systems [1], [2].

F. Limited Feedback Mechanisms

- Some communication devices may not provide immediate visual or tactile confirmation after successful message transmission.
- Users may therefore have difficulty verifying whether the intended communication has been successfully recognized.

G. Need for a Simple Assistive Communication System

- There is a need for a compact, low-cost, portable, and reliable communication system that enables users to communicate predefined messages using minimal hardware.
- The system should provide wireless communication without requiring internet connectivity while ensuring message recognition and user feedback.
- The proposed system addresses these challenges by providing a simple push-button-based communication interface, wireless message transmission through ESP-NOW, OLED-based message display, and vibration feedback for communication confirmation [3], [7], [9].

IV. PROPOSED SOLUTION

The proposed system provides a simple wireless communication platform using two ESP32 microcontrollers configured as transmitter and receiver units. The transmitter generates predefined short and long button-press patterns and transmits them using ESP-NOW. The receiver identifies the received pattern and provides visual and tactile feedback.

A. Transmitter Unit

- The transmitter consists of an ESP32 microcontroller and a push button.
- Short and long button presses are detected according to their duration.
- The detected inputs are represented using **S** for short and **L** for long presses.

B. Pattern Generation

- Individual **S** and **L** inputs are combined to form predefined communication patterns.
- Each predefined pattern corresponds to a message stored in the receiver.
- The generated pattern is prepared for transmission.

C. ESP-NOW Wireless Communication

- The generated pattern is transmitted from the transmitter to the receiver using ESP-NOW.
- ESP-NOW supports direct ESP32-to-ESP32 communication [3].
- Its use in real-time embedded communication has also been investigated in previous work [24].

D. Receiver Unit

- The receiver ESP32 monitors incoming communication data.
- The received pattern is compared with predefined patterns stored in memory.
- A matching pattern is converted into its corresponding message.

E. OLED Display

- The recognized message is displayed on the SSD1306

OLED.

- The SSD1306 controller is suitable for embedded display applications [7].
- The display library provides software support for OLED integration [9].

F. Vibration Feedback

- The vibration motor is activated after successful pattern recognition.
- The vibration provides tactile confirmation to the user.
- An appropriate alert can also be generated for an invalid pattern.

G. Embedded Implementation

- ESP32 provides the main processing and wireless communication platform [1], [2].
- ESP-IDF provides the development framework and software support required for ESP32 applications [4].

H. Communication Operation

- The complete system combines input generation, pattern formation, wireless transmission, pattern recognition, message display, and vibration feedback.
- This integration provides a compact approach to predefined silent communication.

V. METHODOLOGY

The proposed system follows a systematic sequence of operations involving embedded-system initialization, input acquisition, pattern generation, wireless transmission, pattern recognition, and user feedback. The methodology is based on ESP32 embedded technology and wireless communication principles [1], [2], [12]–[14]. The overall implementation is organized into the following stages.

A. System Initialization

- The transmitter and receiver ESP32 microcontrollers are initialized and configured for system operation [1], [2].
- GPIO pins required for the push button, OLED display, and vibration motor are configured.
- The software environment and device interfaces are initialized using the ESP32 development framework [4].
- ESP-NOW communication is established between the two ESP32 modules for direct wireless data exchange [3], [24].
- The predefined communication patterns are stored in the receiver memory.

B. Button Press Detection

- The transmitter continuously monitors the push button for user input.
- The duration of each button press is measured by the ESP32.
- Each input is classified as either Short (S) or Long (L) according to the programmed timing conditions.
- The input-processing operation is implemented using embedded-system programming techniques [4], [10], [12].

C. Communication Pattern Generation

- The detected short and long inputs are combined to generate predefined communication patterns.
- Each generated pattern is associated with a corresponding predefined message.
- The generated pattern is temporarily stored before wireless transmission.
- The use of predefined symbols provides a simple input mechanism for assistive communication applications [18]–[20].

D. Wireless Pattern Transmission

- The generated communication pattern is encoded and transmitted using ESP-NOW.
- The transmitter sends the pattern directly to the receiver ESP32 [3].
- ESP-NOW eliminates the need for a conventional Wi-Fi router or internet connection for communication between the two devices [3], [24].
- Direct wireless communication reduces the requirement for additional network infrastructure and supports compact embedded implementations [16], [21], [23].

E. Pattern Reception

- The receiver ESP32 continuously monitors the wireless communication interface for incoming data.
- After receiving the transmitted data, the communication pattern is extracted from the received information.
- The received pattern is passed to the recognition process for further processing.
- Wireless data exchange between embedded devices forms an important part of distributed embedded communication systems [15], [16], [24].

F. Pattern Recognition

- The receiver compares the received pattern with the predefined patterns stored in its memory.
- If a matching pattern is found, the corresponding predefined message is identified.
- Patterns that do not match the stored patterns are treated as invalid.
- Pattern-based message interpretation provides a simple mechanism for predefined assistive communication [18]–[20].

G. Message Display

- The recognized message is displayed on the SSD1306 OLED display.
- The SSD1306 controller provides the display interface required for presenting information in the embedded system [7].
- The OLED display is integrated using the corresponding software library [9].
- If the received pattern does not correspond to a predefined pattern, the display indicates “**Invalid Pattern.**”

H. Vibration Feedback

- After successful pattern recognition, the receiver

activates the vibration motor.

- The vibration provides tactile confirmation of successful message recognition.
- Feedback mechanisms are important in communication-oriented assistive systems because they provide an indication of system response [18]–[20].
- An appropriate alert vibration is generated when an invalid pattern is detected.

I. Continuous Communication

- After completing one communication cycle, the system returns to the monitoring state.
- The transmitter waits for the next button input, while the receiver continues monitoring for incoming patterns.
- The process is repeated for subsequent communication attempts.
- Continuous operation is supported by the embedded software architecture and real-time processing capabilities of the ESP32 platform [1], [4], [10].

VI. BLOCK DIAGRAM

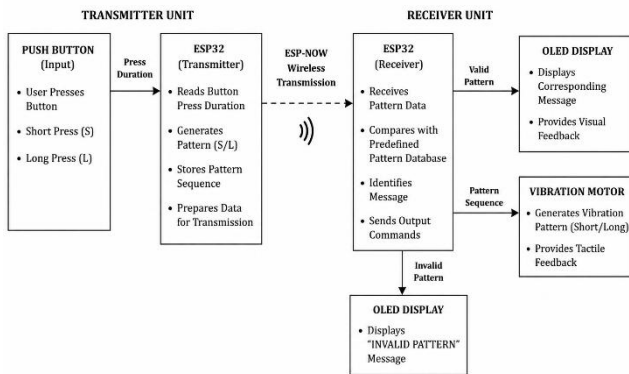


Fig.1. System Architecture of the Proposed Adaptive Silent Communication System

The block diagram represents the complete communication process, consisting of a transmitter unit, wireless communication link, and receiver unit. The system integrates embedded processing, wireless communication, pattern recognition, visual display, and tactile feedback. ESP32 provides the processing and wireless capabilities required for embedded applications [1], [2], while embedded-system design principles support the integration of input and output devices [12], [13].

A. Push Button

- The push button serves as the input device of the transmitter unit.
- The user generates communication patterns by performing short and long button presses.
- The duration of each press is detected and processed by the ESP32 microcontroller [1], [2].

B. ESP32 (Transmitter)

- The transmitter ESP32 measures the button press duration and processes the input.
- The detected inputs are converted into

communication symbols, **S** for short press and **L** for long press.

- These symbols are combined to generate predefined communication patterns.
- The generated pattern is stored and prepared for wireless transmission.
- The use of microcontrollers for input processing and embedded control is consistent with established embedded-system design approaches [10]–[13].

C. ESP-NOW Wireless Communication

- The generated communication pattern is transmitted from the transmitter ESP32 to the receiver ESP32 using ESP-NOW.
- ESP-NOW provides direct wireless communication between ESP32 devices without requiring a conventional Wi-Fi router or internet connection [3].
- The protocol is suitable for direct wireless data exchange between embedded devices [3], [15], [16].
- ESP-NOW-based communication has also been investigated for real-time embedded and wireless applications [3], [24].

D. ESP32 (Receiver)

- The receiver ESP32 continuously monitors incoming communication patterns.
- After receiving the transmitted data, the pattern is extracted and processed.
- The received pattern is compared with the predefined pattern database stored in memory.
- If a valid pattern is identified, the corresponding message is selected for display.
- Processing and communication using ESP32-based embedded platforms are supported by the ESP32 technical documentation [1], [2] and embedded-system principles [12], [13].

E. SSD1306 OLED Display

- The OLED display provides visual feedback by displaying the predefined communication message corresponding to the received pattern.
- The SSD1306 controller provides the interface required for OLED-based embedded display applications [7].
- Software libraries can be used to simplify SSD1306 integration with embedded platforms [9].
- If an invalid communication pattern is received, the display shows “**Invalid Pattern.**”

F. Vibration Motor

- After successful pattern recognition, the vibration motor is activated.
- The vibration confirms successful message reception and provides tactile feedback to the user.
- For an invalid communication pattern, a short vibration alert is generated.
- Tactile feedback provides an additional interaction mechanism for assistive communication systems [18]–[20].

G. Overall System Operation

- The transmitter generates communication patterns using a single push button.
- The generated patterns are transmitted wirelessly through ESP-NOW [3].
- The receiver performs pattern recognition and selects

the corresponding message.

- The recognized message is displayed on the OLED display using the SSD1306 interface [7], [9].
- The vibration motor provides tactile feedback after pattern recognition.
- The integration of wireless communication and embedded processing provides a compact approach to assistive communication [18]–[21].

VII. COMPONENTS REQUIRED

The components required for the implementation of the proposed Adaptive Silent Communication System are classified into Hardware Components, Communication Protocol, and Software Components.

A. HARDWARE SETUP AND COMPONENT ANALYSIS

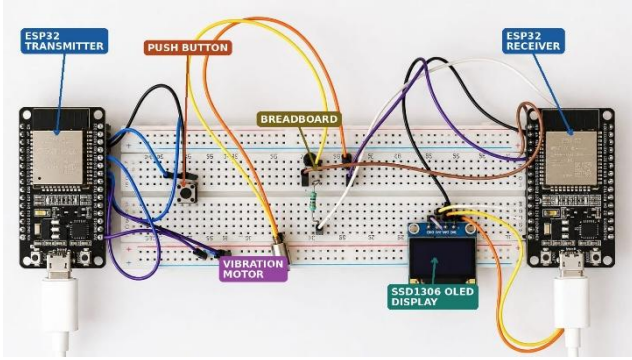


Fig. 2. Integrated Hardware Circuit Setup consisting of (1) ESP32 Transmitter, (2) ESP32 Receiver, (3) Push Button, (4) Vibration Motor, (5) SSD1306 OLED Display, (6) Breadboard and (7) USB Power Connections.

A. ESP32 Transmitter

The ESP32 transmitter acts as the input and communication unit of the proposed system. It receives user input through the push button and processes the duration of each button press. The ESP32 converts short and long button presses into predefined communication symbols and prepares the resulting pattern for wireless transmission [1], [2]. Its integrated wireless capabilities make it suitable for communication between the transmitter and receiver units [1]–[3].

B. ESP32 Receiver

The ESP32 receiver acts as the processing and output-control unit. It receives the transmitted communication pattern through ESP-NOW and compares the received pattern with the predefined patterns stored in memory [1]–[3]. After identifying the pattern, the receiver generates the corresponding output for the OLED display and vibration motor.

C. Push Button

The push button serves as the primary input device of the transmitter. The user generates short and long presses to create the required communication pattern. The ESP32 measures the duration of each press and classifies the input as **Short (S)** or **Long (L)**. The use of simple input devices supports low-complexity embedded-system implementations [12]–[14].

D. ESP-NOW Wireless Communication

ESP-NOW is used as the wireless communication mechanism

between the two ESP32 boards. The generated pattern is transmitted directly from the transmitter to the receiver without requiring a conventional Wi-Fi router or internet connection [3]. ESP-NOW provides a suitable communication mechanism for direct data exchange between ESP32-based embedded devices [3], [24].

E. SSD1306 OLED Display

The SSD1306 OLED display is connected to the receiver ESP32 and provides visual feedback to the user. After successful pattern recognition, the corresponding predefined message is displayed on the OLED. The SSD1306 controller is designed for OLED display applications and is suitable for embedded systems [7]. Supporting display libraries can be used to interface the OLED with the ESP32 [9].

F. Vibration Motor

The vibration motor provides tactile feedback at the receiver side. After a valid communication pattern is recognized, the ESP32 activates the motor according to the corresponding vibration pattern. This provides an additional feedback mechanism along with the visual message displayed on the OLED. Tactile feedback can be useful in assistive communication systems where alternative forms of user interaction are required [18]–[20].

G. Breadboard

The breadboard provides a temporary platform for assembling the electronic components during prototype development. It allows the ESP32 boards, push button, OLED display, vibration motor, and associated connections to be arranged without permanent soldering. This arrangement facilitates testing, debugging, and modification of the prototype during development [12]–[14].

H. Jumper Wires and Power Connections

Jumper wires are used to establish the required power, ground, and signal connections between the ESP32 boards and peripheral components. The transmitter and receiver boards are powered through their USB connections during testing. The wiring arrangement provides the required electrical connections between the microcontrollers, push button, OLED display, and vibration motor.

B. SOFTWARE COMPONENTS

1. Visual Studio Code (VS Code)

Visual Studio Code (VS Code) is used as the Integrated Development Environment (IDE) for developing the proposed system. It provides source-code editing, debugging, project management, and integration with embedded development tools [8]. In this project, VS Code is used for developing, compiling, flashing, and testing the transmitter and receiver firmware.

2. ESP-IDF

The Espressif IoT Development Framework (ESP-IDF) is the official software development framework for ESP32 microcontrollers. It provides APIs, libraries, peripheral drivers, configuration tools, and development support for ESP32-based applications [4]. In the proposed system, ESP-IDF is used to develop the transmitter and receiver firmware, configure GPIO peripherals, implement wireless communication, interface with the OLED display, and control the vibration motor.

C. COMMUNICATION PROTOCOL

ESP-NOW Protocol

ESP-NOW is a wireless communication protocol provided by Espressif for ESP32 devices. It supports direct device-to-device communication without requiring a conventional Wi-Fi router or internet connection [3].

In the proposed system, ESP-NOW is used to transmit the generated communication patterns from the transmitter ESP32 to the receiver ESP32. The transmitter prepares the short and long press pattern and sends it wirelessly to the receiver. The receiver processes the received pattern and performs pattern recognition to determine the corresponding predefined message.

The use of ESP-NOW eliminates the requirement for external network infrastructure and makes it suitable for direct wireless communication between the two ESP32 units [3], [24].

VIII. FLOW CHART

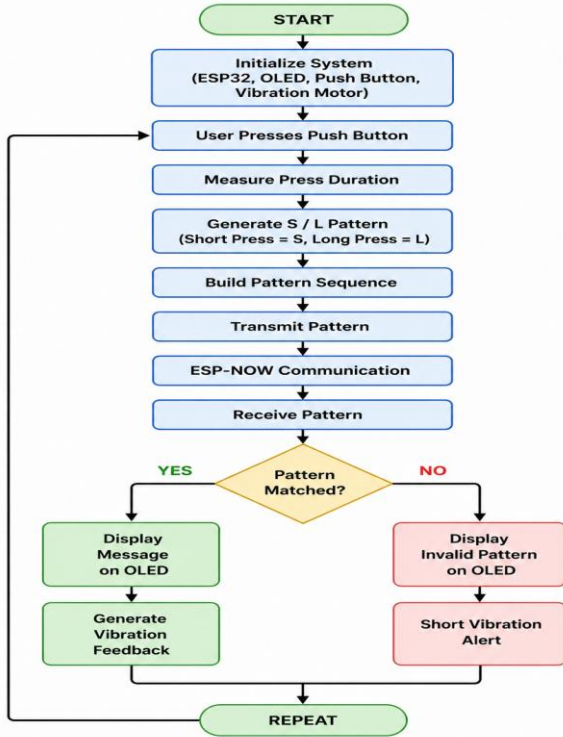


Fig.3. Flowchart of the Proposed System

The flowchart shown in Fig. 3 illustrates the complete operational sequence of this project, including system initialization, user input, pattern generation, wireless transmission, pattern recognition, message display, and vibration feedback. The ESP32 provides the embedded processing platform, while ESP-IDF provides the software framework and peripheral interfaces required for implementation [1], [2], [4].

A. Start

The execution begins by powering on the transmitter and receiver units. The ESP32-based embedded platform provides the required processing environment for the system [1], [4].

B. System Initialization

The ESP32 microcontrollers initialize the required hardware components, including the push button, SSD1306 OLED display, vibration motor, and wireless communication functions. ESP-IDF provides the APIs and peripheral drivers required for configuring and controlling these components [1], [4], [12].

The receiver also loads the predefined communication patterns into memory for subsequent pattern recognition.

C. User Input

The user presses the push button to generate a communication pattern. The transmitter continuously monitors the button input through the ESP32 GPIO interface. GPIO-based input monitoring is commonly used in embedded-system applications [1], [4], [12].

D. Press Duration Measurement

The transmitter measures the duration of each button press using the ESP32 timing functions. A short press is represented by S, while a long press is represented by L. Timing and event handling are implemented using the embedded software facilities provided by ESP-IDF [4], [12], [13].

E. Pattern Generation

Based on the measured button-press duration, the transmitter generates the corresponding communication symbol. The detected symbols are combined to form the required communication pattern. The pattern-generation and encoding logic is implemented as part of the software developed for this project.

F. Pattern Sequence Formation

The generated symbols are arranged into a complete pattern sequence, such as S, SL, SLS, or LSS, according to the user's input. The resulting sequence is temporarily stored before transmission.

G. Pattern Transmission

After the complete pattern is generated, the transmitter prepares the data packet for wireless transmission. ESP32 provides the wireless communication capabilities required for communication between the two embedded units [1], [2].

H. ESP-NOW Communication

The communication pattern is transmitted from the transmitter ESP32 to the receiver ESP32 using ESP-NOW. ESP-NOW supports direct device-to-device communication between ESP32 devices without requiring a conventional Wi-Fi router or internet connection [3], [4].

The use of direct wireless communication reduces the dependency on external network infrastructure and is suitable for embedded communication applications [3], [24].

I. Pattern Reception

The receiver ESP32 receives the transmitted communication pattern through the ESP-NOW interface. The received data is then passed to the pattern-recognition process for further processing [3], [4].

J. Pattern Matching

The received communication pattern is compared with the predefined pattern database stored in the receiver memory.

- If the received pattern matches a predefined pattern, the corresponding message is identified.

- If no matching pattern is found, the received sequence is classified as an invalid communication pattern. The pattern-matching operation is implemented using the software logic developed for this project.

K. Message Display

For a valid communication pattern, the corresponding predefined message is displayed on the SSD1306 OLED display. The SSD1306 controller provides the required interface for OLED-based embedded display applications [7]. The OLED is interfaced with the ESP32 through the appropriate software driver/library [9].

L. Vibration Feedback

After successful pattern recognition, the receiver activates the vibration motor to provide tactile feedback and confirm successful message reception. Tactile feedback can provide an additional interaction mechanism in assistive communication systems [18]–[20].

M. Invalid Pattern Handling

If the received pattern does not match any predefined pattern, the OLED displays “Invalid Pattern” [7], [9]. The vibration motor then generates a short alert to indicate that the received sequence could not be recognized.

N. Repeat

After completing one communication cycle, the system returns to the monitoring state. The transmitter waits for the next button input while the receiver continues to monitor for incoming patterns. Continuous embedded processing and wireless communication allow the system to repeat the communication process [3], [4], [12].

X. WORKING PRINCIPLE

The proposed system enables silent wireless communication between two ESP32 units, with one acting as the transmitter and the other as the receiver. The ESP32 provides the required processing, GPIO, and wireless communication capabilities [1], [3].

Initially, both units are initialized. The transmitter configures the push button, while the receiver initializes the SSD1306 OLED, vibration motor, and wireless communication functions using the ESP-IDF framework [3], [4].

The user generates a communication pattern using short and long button presses. The transmitter measures the press duration and converts it into S (Short) or L (Long) symbols. These symbols are combined to form predefined patterns such as S, L, SL, SLS, and LSS [1], [3].

The generated pattern is transmitted from the transmitter to the receiver using ESP-NOW. ESP-NOW supports direct communication between ESP32 devices without requiring a Wi-Fi router or internet connection [2], [5].

The receiver compares the received pattern with the predefined pattern database. For a valid pattern, the corresponding message is displayed on the SSD1306 OLED [4], and the vibration motor provides tactile feedback. Assistive communication systems can benefit from alternative visual and tactile feedback mechanisms [7]–[9].

For an invalid pattern, the OLED displays “Invalid Pattern” and the vibration motor generates a short alert. The system then returns to the monitoring state and waits for the next input, enabling continuous silent communication [1]–[3].

IX. RESULTS

```
I (650) wifi_init: WiFi RX IRAM OP enabled
I (650) phy_init: phy_version 4791,2c4672b,Dec 20 2023,16:06:06
I (750) wifi:mode : sta (28:05:a5:2f:b4:e0)
I (750) wifi:enable tsf
I (750) ESPNOW: espnow [version: 1.0] init
TRANSMITTER READY
Pattern: S
Pattern: SL
Pattern: SLS
Sent Pattern: SLS
```

Fig. 4. Serial Monitor Output of the Transmitter.

```
I (716) wifi_init: WiFi RX IRAM OP enabled
I (716) phy_init: phy_version 4791,2c4672b,Dec 20 2023,16:06:06
I (806) wifi:mode : sta (30:76:f5:f4:29:44)
I (816) wifi:enable tsf
I (816) ESPNOW: espnow [version: 1.0] init
Receiver Ready
Received: S
Received: L
Received: SLS
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Fig. 5. Serial Monitor Output of the Receiver.

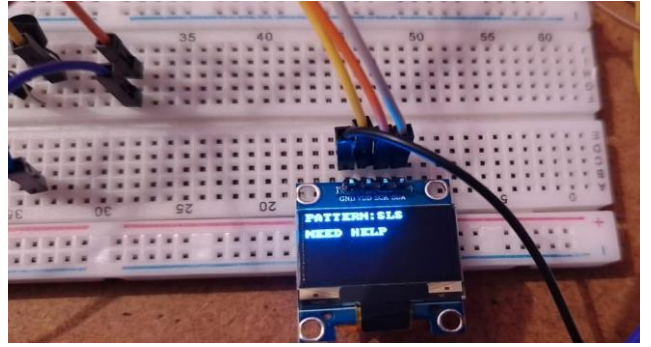


Fig. 6. OLED Display Showing the Recognized Pattern and Corresponding Message.

A. Transmitter Output

Fig. 4 shows the serial monitor output of the transmitter ESP32 during testing. The transmitter successfully initializes the ESP-NOW protocol and indicates “TRANSMITTER READY.” The system detects the user's button presses and generates the communication patterns S, SL, and SLS. The output also shows “Sent Pattern: SLS,” confirming that the generated pattern was successfully prepared and transmitted.

B. Receiver Output

Fig. 5 shows the serial monitor output of the receiver ESP32. The receiver successfully initializes the ESP-NOW protocol and indicates “Receiver Ready.” The transmitted patterns S, L, and SLS are successfully received and displayed in the serial monitor. This confirms successful wireless communication and data reception between the two ESP32 units.

C. OLED Display Output

Fig. 6 shows the SSD1306 OLED display after successful pattern recognition. The received pattern SLS is displayed as “PATTERN: SLS”, and the corresponding predefined message “NEED HELP” is displayed below it. This confirms that the receiver successfully matches the received pattern with the predefined pattern and generates the corresponding communication message.

D. Overall Result

The experimental results demonstrate the successful operation of the proposed system from pattern generation to message display. The transmitter generates and sends the communication pattern, the receiver successfully receives it, and the OLED displays the corresponding predefined message.

The results confirm the functional integration of the ESP32 transmitter, ESP-NOW communication, receiver, pattern recognition, and OLED display.

XI. FUTURE SCOPE

The developed system provides a basic platform for silent wireless communication using predefined vibration patterns. The following improvements can be considered in future development:

Expansion of Communication Patterns: More short- and long-press combinations can be added to support a larger number of predefined messages.

Adaptive Pattern Recognition: Intelligent pattern-recognition techniques can be incorporated to improve recognition accuracy and handle variations in button-press duration.

Bidirectional Communication: Future versions can support communication in both directions, allowing both users to send and receive messages.

Improved Feedback Mechanism: Multiple vibration patterns and improved OLED interfaces can be implemented to provide clearer tactile and visual feedback.

Compact Hardware Design: The prototype can be integrated into a smaller and portable enclosure for convenient use in real-world environments.

Enhanced Wireless Range: Future implementations can investigate suitable wireless communication techniques to improve communication range and reliability.

Mobile or Cloud Integration: A mobile application or optional cloud platform can be incorporated for message logging, monitoring, and remote access when network connectivity is available.

Emergency Communication: Dedicated emergency patterns can be introduced for quickly transmitting predefined alerts such as help, medical assistance, or emergency notifications.

XII. CONCLUSION

This paper presents the development of a compact silent communication system based on ESP32 microcontrollers, predefined short- and long-press patterns, and ESP-NOW wireless communication. The proposed system enables a user to generate communication patterns through a simple push-button interface, transmit the patterns wirelessly, and identify the corresponding messages at the receiver.

The implemented prototype integrates pattern generation, ESP-NOW-based wireless transmission, pattern recognition, SSD1306 OLED display, and vibration feedback into a single communication framework. Experimental testing demonstrated successful generation and reception of predefined patterns, including **S**, **L**, and **SLS**. The successful recognition of the **SLS** pattern and display of the corresponding **"NEED HELP"** message on the OLED confirm the functional operation of the developed system.

The results demonstrate that the proposed approach can provide a simple and practical mechanism for silent communication without depending on internet connectivity. The developed prototype establishes a foundation for further improvements in pattern recognition, communication capacity, portability, and feedback mechanisms for assistive

communication applications.

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